

Homework 2

EE 999 — Introduction to Control Systems

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Due Wednesday, September 23, 2026

Problem 1.A.2.1 — What is a "free-body diagram"?

A free body diagram is a diagram that shows the forces acting on an object or set of objects.

Problem 1.A.2.2 — What are the two forms of Newton's law?

Translational motion: $F = ma$; Rotational motion: $M = I\alpha$

Problem 1.A.2.3 — For a structural process to be controlled, such as a robot arm, what is the meaning of "collocated control" and "Noncollocated control"?

Collocated control means the sensor and actuator are rigidly attached to one another, whereas there is flexibility between the actuator and sensor in a noncollocated system.

Problem 1.A.2.4 — State Kirchhoff's current law.

Kirchhoff's current law states that current entering any given electrical node must equal current exiting a node.

Problem 1.A.2.5 — State Kirchhoff's voltage law.

Kirchhoff's voltage law states that the sum of all voltages around a closed circuit must equal zero.

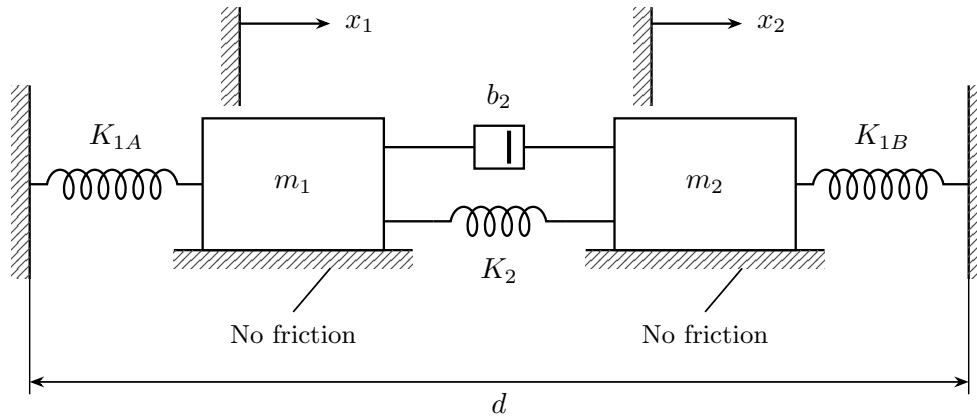
Problem 1.B — Differential equation for mechanical system

Figure 1: The two-mass system of Problem 1.B.

Signal	Units	Description
$x_1(t)$	[m]	Position of mass m_1
$x_2(t)$	[m]	Position of mass m_2

Table 1: Signals of Problem 1.B.

Parameter	Units	Description
m_1	[Kg]	mass of m_1
m_2	[Kg]	mass of m_2
K_{1A}	[N/m]	Outer spring stiffness, wall to m_1
K_{1B}	[N/m]	Outer spring stiffness, m_2 to wall
K_2	[N/m]	Inner spring stiffness
b_2	[N-s/m]	Shock absorber

Table 2: Parameters of Problem 1.B.

Constituent equations:

$$\begin{aligned}
 f_{K_{1A}}(t) &= K_{1A}x_1(t) \\
 f_{K_{1B}}(t) &= K_{1B}(-x_2(t)) \\
 f_{K_2}(t) &= K_2(x_2(t) - x_1(t)) \\
 f_{b_2} &= b_2(\dot{x}_2(t) - \dot{x}_1(t)) \\
 f_{net_{m1}}(t) &= m_1\ddot{x}_1(t) \\
 f_{net_{m2}}(t) &= m_2\ddot{x}_2(t)
 \end{aligned}$$

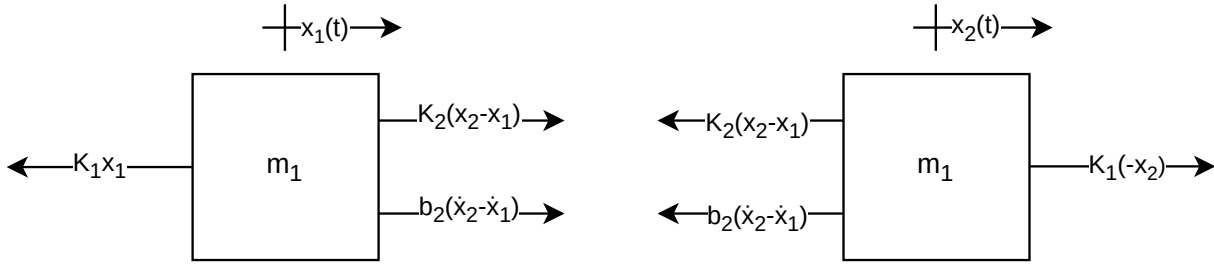


Figure 2: Free body diagram for problem 1.B

Continuity Equations:

$$K_1 = K_{1A} = K_{1B}$$

$$m = m_1 = m_2$$

$$\begin{aligned} m\ddot{x}_1(t) &= f_{K_2} + f_{b_2} - f_{K_{1A}} \\ &= K_2(x_2 - x_1) + b_2(\dot{x}_2 - \dot{x}_1) - K_1x_1 \\ &= K_2x_2 - K_2x_1 + b_2\dot{x}_2 - b_2\dot{x}_1 - K_1x_1 \\ &= -b_2\dot{x}_1 - (K_1 + K_2)x_1 + b_2\dot{x}_2 + K_2x_2 \end{aligned}$$

$$\begin{aligned} m\ddot{x}_2(t) &= f_{K_{1B}} - f_{K_2} - f_{b_2} \\ &= K_1(-x_2) - K_2(x_2 - x_1) - b_2(\dot{x}_2 - \dot{x}_1) \\ &= -K_1x_2 - K_2x_2 + K_2x_1 - b_2\dot{x}_2 + b_2\dot{x}_1 \\ &= -b_2\dot{x}_2 - (K_1 + K_2)x_2 + b_2\dot{x}_1 + K_2x_1 \end{aligned}$$

Standard form:

$$m\ddot{x}_1(t) + b_2\dot{x}_1(t) + (K_1 + K_2)x_1(t) = b_2\dot{x}_2(t) + K_2x_2(t) + f(t)$$

$$m\ddot{x}_2(t) + b_2\dot{x}_2(t) + (K_1 + K_2)x_2(t) = b_2\dot{x}_1(t) + K_2x_1(t)$$

*NOTE: $f(t)$ has been added to the m_1 equation so the transfer function could be created

Frequency domain:

$$(ms^2 + b_2s + (K_1 + K_2))X_1(s) = (b_2s + K_2)X_2(s) + F(s)$$

$$(ms^2 + b_2s + (K_1 + K_2))X_2(s) = (b_2s + K_2)X_1(s)$$

Find $X_1(s)$ in terms of $X_2(s)$ so substitution may be performed:

$$X_1(s) = \frac{ms^2 + b_2s + (K_1 + K_2)}{b_2s + K_2}X_2(s)$$

Then perform the substitution:

$$\begin{aligned}
 (ms^2 + b_2s + (K_1 + K_2)) \frac{ms^2 + b_2s + K_1 + K_2}{b_2s + K_2} X_2(s) &= (b_2s + K_2)X_2(s) + F(s) \\
 (ms^2 + b_2s + (K_1 + K_2))^2 X_2(s) &= (b_2s + K_2)^2 X_2(s) + (b_2s + K_2)F(s) \\
 ((ms^2 + b_2s + (K_1 + K_2))^2 - (b_2s + K_2)^2) X_2(s) &= (b_2s + K_2)F(s)
 \end{aligned}$$

Next, gather the terms:

s^4	s^3	s^2	s^1	s^0
m^2	mb_2	$m(K_1 + K_2)$		
	mb_2	b_2^2	$b_2(K_1 + K_2)$	
		$m(K_1 + K_2)$	$b_2(K_1 + K_2)$	$K_1^2 + 2K_1K_2 + K_2^2$
		$-b_2^2$	$-b_2K_2$	
			$-b_2K_2$	$-K_2^2$

Then, combine them to write the entire transfer:

$$G_{FX_2(s)} = \frac{X_2(s)}{F(s)} = \frac{b_2s + K_2}{m^2s^4 + 2mb_2s^3 + 2m(K_1 + K_2)s^2 + (2b_2(K_1 + K_2) - 2b_2K_2)s + K_1^2 + 2K_1K_2}$$

Check Units:

$$\begin{aligned}
 &\frac{\frac{Ns}{m}s^{-1} + \frac{N}{m}}{Kg^2s^{-4} + Kg\frac{Ns}{m}s^{-3} + Kg\frac{N}{m}s^{-2} + \frac{Ns}{m}\frac{N}{m}s^{-1} + (\frac{N}{m})^2} \\
 &\frac{\frac{N}{m}s^{-1} + \frac{N}{m}}{\frac{N^2}{m^2}s^{-4} + \frac{N}{m}\frac{N}{m}s^{-3} + \frac{N}{m}\frac{N}{m}s^{-2} + \frac{N}{m}\frac{N}{m}s^{-1} + (\frac{N}{m})^2} \\
 &\frac{\frac{N}{m}}{\frac{N^2}{m^2}} = \frac{m}{N}
 \end{aligned}$$

This makes sense because the input is a force and the output is a displacement, so the units of the transfer function should be distance over force.

The difference in velocities between m_1 and m_2 will decay because b_2 scales with velocity. Since there is no friction, the output $x_2(t)$ will not decay.

Problem 1.C — Modified cruise control

Equation 2.4 from Example 2.1 in the text book reads:

$$\dot{v} + \frac{b}{m}v = \frac{u}{m}$$

Substituting $u = K(v_r - v)$, the equation becomes:

$$\dot{v} + \frac{b}{m}v = \frac{K(v_r - v)}{m}$$

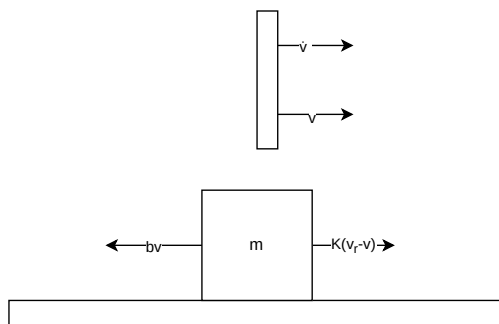


Figure 3: Schematic diagram of car cruise control model

Signal	Units	Description
$v(t)$	[m]	Measured velocity of car
$v_r(t)$	[m]	Commanded velocity of car

Table 3: Signals of Problem 1.B.

Parameter	Units	Description
m	[Kg]	mass of car
b	[N-s/m]	Outer spring stiffness, wall to m
K	[N-s/m]	gain constant

Table 4: Parameters of Problem 1.B.

constituent equations:

$$\begin{aligned} f_b &= bv \\ f_K &= K(v_r - v) \\ f_{net} &= m\dot{v} \end{aligned}$$

Continuity equations:

$$\begin{aligned} f_{net} &= f_K - f_b \\ m\dot{v}(t) &= K(v_r(t) - v(t)) - bv(t) \end{aligned}$$

Standard form:

$$Kv_r(t) = m\dot{v}(t) + Kv(t) + bv(t)$$

$$Kv_r(t) = m\dot{v}(t) + (K + b)v(t)$$

Frequency domain:

$$KV_r(s) = (ms + (K + b))V(s)$$

Transfer function:

$$G_{V(s)} = \frac{V(s)}{V_r(s)} = \frac{K}{ms + (K + b)}$$

Units:

$$\frac{\frac{N \cdot s}{m}}{Kg \cdot s^{-1} + \frac{N \cdot s}{m} + \frac{N \cdot s}{m}}$$

$$\frac{\frac{N \cdot s}{m}}{\frac{N \cdot s}{m} \cdot s^{-1} + \frac{N \cdot s}{m} + \frac{N \cdot s}{m}}$$